must implode. The next challenge was obvious: use the laws of physics to compute the details of the implosion. What would the implosion look like as seen by people in orbit around the star? What would it look like as seen by people riding on the star's surface? What would be the final state of the imploded star, thousands of years after the implosion?

This computation would not be easy. Its mathematical manipulations would be the most challenging that Oppenheimer and his students had yet tackled: The imploding star would change its properties rapidly as time passes, whereas the Oppenheimer-Volkoff neutron stars had been static, unchanging. Spacetime curvature would become enormous inside the imploding star, whereas it had been much more modest in neutron stars. To deal with these complexities would require a very special student. The choice was obvious: Hartland Snyder.

Snyder was different from Oppenheimer's other students. The others came from middle-class families; Snyder gave the impression of working class, though his father was actually an engineer. It was rumored (incorrectly) that he was a truck driver in Utah before turning physicist. As Robert Serber recalls, "Hartland pooh-poohed a lot of things that were standard for Oppie's students, like appreciating Bach and Mozart and going to string quartets and liking fine food and liberal politics."

The Caltech nuclear physicists were a more rowdy bunch than Oppenheimer's entourage; on Oppenheimer's annual spring trek to Pasadena, Hartland fit right in. Says Caltech's William Fowler, "Oppie was extremely cultured; knew literature, art, music, Sanskrit. But Hartland—he was like the rest of us bums. He loved the Kellogg Lab parties, where Tommy Lauritsen played the piano and Charlie Lauritsen [leader of the lab] played the fiddle and we sang college songs and drinking songs. Of all of Oppie's students, Hartland was the most independent."

Hartland was also different mentally. "Hartland had more talent for difficult mathematics than the rest of us," recalls Serber. "He was very good at improving the cruder calculations that the rest of us did." It was this talent that made him a natural for the implosion calculation.

Before embarking on the full, complicated calculation, Oppenheimer insisted (as always) on making a first, quick survey of the problem. How much could be learned with only a little effort? The key to this first survey was Schwarzschild's geometry for the curved space-time outside a star (Chapter 3).